Edward Ajayi

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Education

Carnegie Mellon University
Master of Science in Engineering Artificial Intelligence — *Aug. 2024 – Present*
- **Relevant Courses:** Natural Language Processing, Deep Learning, Question Answering with LLMs
- **Specializations:** Data Science, Machine Learning, NLP
- **GPA:** 3.77/4.0

Ekiti State University
Bachelor of Science in Computer Science — *Oct. 2014 – Nov. 2019*
- **Specializations:** Computer Science
- **GPA:** 4.17/5.0

Research Interests

Socially Aware AI, Multimodal Reasoning, Computational Humor, Culturally Grounded NLP, AI for Healthcare

Selected Publications

1. **E. Ajayi**, M. Kachweka, M. Deku, and E. Aiken. *A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media. Proceedings of Machine Learning Research (PMLR) / AAAI-26 AIMedHealth Bridge* (Oral). [Paper Link]
2. **E. Ajayi** and P. Mitra. *Automatic Humor Detection: From Theoretical Foundations to Large Language Models.* To be submitted to *ACM Computing Surveys*. [Manuscript Link]
3. **E. Ajayi**, B. Tadele, E. Umwari, M. Deku, P. Singadi, C. Edeh, and J. Udahemuka. *Language Diversity: Evaluating AI Performance on African Languages.* [Manuscript Link]

Experience

Research Assistant
Spatial and Language Technologies Lab, CMU — *Aug. 2025 – Dec. 2025*
- Conducting research on NLP for automatic humour detection and generation, dataset preparation, model development, and evaluation.
- Contributing to projects on low-resource language technologies and advancing cultural understanding in large language models.

Graduate Researcher
Carnegie Mellon University — *Jan. 2025 – Present*
- Researched machine learning methods for detecting mental health conditions and cyberbullying from social media, resulting in a paper accepted at AAAI-26.
- Conducting two ongoing research projects in Question Answering with Large Language Models, focusing on reasoning, evaluation, and robustness.

Research Intern - Business Automation
Carnegie Mellon University — *Jun. 2025 – Aug. 2025*
- Developing and prototyping an agentic AI framework to automate complex business workflows.
- Researching and implementing AI safety protocols and robust LLM reasoning for ethical alignment and deployment.
- Analyzing agent performance to establish benchmarks for human-in-the-loop oversight and develop explainability metrics to support transparent corporate accountability.

Client Service Officer
Stanbic IBTC Bank — *Dec. 2021 – Jul. 2024*
- Leveraged data-driven insights to cross-sell over N400M in assets, enhancing client trust and digital adoption by 25%.
- Managed client relationships, identifying service gaps to improve operational efficiency.

Undergraduate Researcher
Ekiti State University — *Sep. 2018 – May 2019*
- Developed a decision analyzer enabling robots to evaluate ambiguous and uncertain scenarios.
- Co-authored a paper published in the European Journal of Engineering and Technology.

Technical Skills

Programming & Scripting: Python, SQL, Bash
Data Science & ML: Pandas, NumPy, Scikit-Learn, PyTorch, Matplotlib, Seaborn, Hugging Face Transformers
Natural Language Processing: Langraph, OpenAI Agents SDK, NLTK, spaCy, Sentiment Analysis, LLM Fine-Tuning, Prompt Engineering, Retrieval-Augmented Generation (RAG)
Data Engineering & Tools: Jupyter, Neo4j, FastAPI, BeautifulSoup, Google Sheets API, MS Excel
Cloud & Deployment: AWS, Docker, REST APIs

Projects

Multilabel Machine Learning System for Mental Health and Cyberbullying Detection

- Built a full ML pipeline for multilabel detection of depression, anxiety, PTSD, and cyberbullying using classical models and transformer-based embeddings.
- Produced results leading to an *Oral* paper acceptance at AAAI-26 AIMedHealth. [arXiv]

MyTorch – A Minimal Deep Learning Framework Built from Scratch | *Python*

- Implemented core tensor ops, autograd, activation layers, and SGD without deep learning libraries.
- Designed a PyTorch-style API enabling end-to-end training and illustrating neural network internals.

AI Workflow Automation Agent | *Python, LangChain, FastAPI, Google Sheets API, SMTP*

- Built an autonomous agent that processes Google Form submissions, categorizes requests, and triggers approval workflows.
- Automated notifications and integrated a QA agent that validates updates and writes results in real time.

Training & Conferences

Participant, Deep Learning Indaba 2025 Rwanda
Deep Learning Indaba *Aug. 2025*

- Attended workshops on deep learning, AI for social good, and research collaboration across African universities.

Participant, Digital Humanities Summer School Rwanda
DIGIHUM/CMU-Africa *Jul. 2025*

- Engaged in computational methods for humanities research, including NLP applications.

Volunteering/Leadership

President, CMU-Africa Research Club
Carnegie Mellon University *Jan. 2025 – Dec. 2025*

- Led 12 major events in 2025 supporting student research.
- Launched website with research guides, conference updates, and collaboration hub.
- Organized first Student Research Showcase with 20+ posters and workshops.
- Hosted three-part PhD Application Series, faculty sessions, and alumni panels supporting 40+ students.
- Collaborated with ACM SIGCOMM and Upanzi Network for remote conference participation.
- Built an active research ecosystem that provided critical peer support, contributing to over 20 student acceptances to top-tier venues including NeurIPS and AAAI in 2025.

Reviewer, AAAI AIMedHealth Bridge 2026
AAAI Symposium on AI for Medicine and Health *Nov. 2025 – Nov 2025*

- Reviewed submissions in applied machine learning and AI for healthcare under the AAAI AIMedHealth Bridge 2026 program.

IEEE Student Member, CMU-Africa
Institute of Electrical and Electronics Engineers *Feb. 2025 – Present*

- Collaborating with IEEE members to advance research initiatives among club members.

Certifications

- Microsoft Certified Azure Data Scientist Associate - View
- DataCamp Python Data Associate - View

Honors and Awards

- Mastercard Foundation Scholarship, Carnegie Mellon University Africa
- Federal Scholarship Board Award for Undergraduates, Ekiti State University